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*Office Hours:*

**Class Hours:**

|     |                    |           |
|-----|--------------------|-----------|
| MWF | 11:00 pm -12:00 pm | BUR 220   |
| F   | 1:00 pm-2:00 pm    | CPE 2.220 |

**Prerequisites:** ChE 353 with at least a C

**Text:** “Unit Operations of Chemical Engineering”, 7th Ed.  
McCabe, Smith, and Harriott, New York, McGraw-Hill, 2004 (“MSH”)

**Grading System:**

- |              |      |
|--------------|------|
| • Homework   | 10 % |
| • Quizzes    | 15 % |
| • Exams (3)  | 45 % |
| • Final Exam | 30 % |

Final course grades will be assigned according to the “gap system”. In other words, the final grade distribution for all students will be plotted and cut-offs for grades will be determined based on breaks in the distribution. Thus, a student’s performance in the class is measured relative to the performance of the remainder of the class.

**Course Overview:**

1. Static fluid phenomena
2. Flow of incompressible fluids
3. Flow of compressible fluids
4. Flow past objects
5. Conductive heat transfer
6. Convective heat transfer
7. Heat transfer equipment design
8. Radiation

**Goals:** The course covers analysis and design of heat exchangers, fluid flow equipment and some interphase contacting devices. This course serves to integrate the fundamental concepts learned in ChE 353 with practical chemical engineering problems and industrial applications. Key objectives of this course are to enhance problem definition and problem solving skills of the student, and to improve the student's ability to make calculations and obtain correct final answers.

**Knowledge, Abilities, and Skills Students Should Have Entering This Course:**

1. The catalog lists ChE 353 as a prerequisite. The course relies heavily on the mass, momentum, heat and energy balance equations developed in 353. Some computer programming experience from ChE 210 is valuable.
2. Basic mathematical and computer programming skills.
3. Ability to perform macroscopic mass and energy balances under steady state and dynamic conditions (from ChE 353).
4. Familiarity with equations of state, particularly the ideal gas law (from ChE 317).
5. Familiarity with concepts of enthalpy, work, heat and other thermodynamic quantities (from ChE 317).
6. Mastery of the fundamentals of transport phenomena including familiarity with the equations used to describe momentum and heat transfer, an understanding of dimensionless groups, and the ability to derive momentum and heat balance equations (from ChE 353).

**Knowledge, Abilities, and Skills Students Should Gain From This Course:**

1. Familiarity with the methods used to analyze and design fluid processing equipment, heat exchangers and interphase contacting devices for incompressible fluids.
2. Recognition of the challenges of designing processing equipment for compressible fluids.
3. The ability to analyze and size piping networks, valves, pumps, compressors and flow meters.
4. The ability to determine heat transfer coefficients and the ability to size heat exchangers.
5. The ability to apply the principles of transport phenomena to the analysis of a wide range of processing equipment including evaporators, fluidized beds, and chemical reactors.

**Impact on Subsequent Courses in Curriculum:**

The material taught in ChE 354 will be used extensively in the plant design course and in the unit operations lab. (ChE 264). The material will also be of use in the reactor design course (ChE 372) when topics such as packed and fluidized bed reactors are discussed and in the plant design course (ChE 473K).

### **Course Format & Learning Activities:**

- The supplemental notes provide overall excellent coverage of the application of momentum, heat and mass transfer. These readings are intended to provide you with some theoretical coverage of the field, but will typically emphasize the practical application of unit operation principles.
- Lectures will complement readings from your textbook and class handouts. In general, lectures will be interactive, combining in-class discussions with small group problem solving exercises, chalk board problem analyses, etc. Open-ended questions, similar to problems you may encounter in your later engineering practice, will provide you with opportunities to enhance your problem formulation and problem solving skills. Videos on how certain pieces of equipment function and a field trip to the UT power plant will provide you with the ability to identify pieces of equipment and will also help you to link the theory with the physical.
- Recitation sections will be overseen by the TA and/or the Professor and will focus on the solution of additional relevant problems. In some cases, we will cover difficult homework problems and go through exam solutions during recitation. In all cases, an emphasis will be placed on how a problem is approached and why a particular approach is selected.
- Regular homework assignments on the technical material will be given to provide practice at applying the concepts covered in lecture. Group discussion of the homework (but not copying) is encouraged. However, it is important for exams that each student know how to independently work the assigned problems on his/her own.
- Quizzes will be given in-class at random times throughout the semester and are designed to emphasize lecture and homework concepts.
- Three exams and a final will test both retention of concepts and facts, and the ability to apply problem-solving skills. Material tested will be extracted from the readings, homework, lectures, projects, and recitations.

### **Performance Feedback:**

Feedback on your performance throughout the semester is a key component to the learning process. In addition to feedback on homework, quizzes, and exams through solution sets, comments and grades, you will receive immediate input through in-class problem-solving and group activities. Feedback will also be actively encouraged through visits during hours or by appointment. However, I encourage you to attempt all problems on your own prior to seeking additional help.

## **Course Policies:**

Attendance -- Attendance at lectures and recitation sessions, although not mandatory, is important for your mastery of the subject matter.

## **Homework & Projects --**

Homework and projects must be received at the beginning of the class period in which they are due.

Homework must follow the guidelines from ChE 317: engineering paper must be used, problem statement must be defined, sources of information (Appendices, etc.) must be identified, and an arrow must point to a single boxed solution. Illegible papers or those not adhering to the criteria established in ChE 317 will not be graded. Solutions to all problems must include adequate steps, and explanations where necessary, for us to understand how you arrived at an answer. Just an equation and a final answer will not be acceptable. The total homework score is worth 10 % of your overall course grade.

Grading Policies -- Homework, quizzes, and exams will be **approximately** graded according to the following:

- |     |                                                                                                                                                                                                                                                                                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5%  | Participation (making any honest attempt)                                                                                                                                                                                                                                       |
| 15% | Accurately defining the problem (drawing a correct schematic/interpretation of the system)                                                                                                                                                                                      |
| 40% | Setting up the problem (making accurate assumptions, defining the correct equations, adequately explaining all steps required to reach a solution, justifying your approach)                                                                                                    |
| 30% | Correctness and completeness (making accurate calculations, using correct conversions, using appropriate physical properties, identifying sources of information, accurately interpreting data from tables, carrying solution to completion, using correct significant figures) |
| 10% | Neatness, organization (logical flow), and boxing final solutions                                                                                                                                                                                                               |

Problems should be worked in units that are consistent with the units given in the problem statement. The final solution must be reported in units consistent with those given in the problem statement. For example, if the problem statement gives metric units, do not report your solution in English units.

*Note: Conceptual errors are weighted more heavily than simple mathematical (calculation) errors.*

Late Assignments – No late assignments will be accepted without permission from Dr. Eldridge.

Grading Disputes -- Discussion on assignments should be initiated with the TA or grader, and only math errors and oversights will be considered valid reasons for dispute. If your work is not clear or the specific question involves subjectivity, then there is no justification for re-grading. Grades will not be discussed after one week from the date the assignment is returned.

Quizzes - Quizzes will be **closed book** with required tables and figures supplied. No make-up quizzes will be given. Students missing a quiz with an approved excuse will have the missed quiz dropped from the final grade tabulation.

Exams --

Exams will be **closed book and closed notes** with required tables and figures supplied.. A review session will be held prior to each exam.

Requests for re-grading must be submitted to the professor **within 1 week** after the graded exam has been returned. If re-grading is desired, then the entire exam is subject to re-grading. As with assignments, if your work is not clear or the specific question involves subjectivity, then there is no justification for re-grading.

Missed Exam -- No make-up exams will be given without a written excuse for the absence.

Final Exam -- The location will be announced later in the semester. The date and time of the exam cannot be changed, and there will be no make-up final. Registration for this course includes the university-scheduled final exam date.

Cheating -- Cheating is strictly forbidden and anyone found doing so will be turned over to the University and will be dealt with according to University policy. Working together on homework is encouraged, but each person must independently write-up his or her own work. Copying homework and photocopying graphs and figures between individuals is considered cheating.

Drop Policy– Registration changes after the official drop date require approval by the Department Chair and/or the Dean of Engineering.

Disabilities -- The University of Texas at Austin provides upon request appropriate academic adjustments for qualified students with disabilities. For more information, contact the Office of the Dean of Students at 471-6259, 471-4641 TTY or the College of Engineering Director of Students with Disabilities at 471-4321.